Methodology:

The first aspect taken into consideration for pitcher role reversals was to eliminate pitchers who are already successful in their current roles. The goal of this project is to identify certain pitchers that may be more successful in a different role, meaning current successful pitchers should be left alone. Successful starting pitchers (SPs) are defined as having a 2023 xERA < 4.35, successful relief pitchers (RPs) are defined as having a 2023 xERA <4.50, and successful closing pitchers are defined as having a 2023 xERA <4.00.

The following analysis heavily weighs 3 already established baseball models in xERA, Stuff+, and Location +. Stuff+ metric was not considered for fast balls (due to decreasing velocity as game progresses - would be an unfair comparison between CL and SPs), but were for each of the other 7 Stuff+ pitches.

The next step was to identify characteristics successful pitchers in their specific roles share. It was noticed that a vast majority of successful SPs contain a minimum three quality pitches; defined as a pitch having a Stuff+ rating over 100. RPs have a minimum of two quality pitches, and closers have at least one elite pitch (in the 95th percentile of Stuff+) or 2 quality pitches to be successful in their role. Other factors such as age, injury history, BB/9, K/9, Location +, and split stats were used to evaluate the leading candidates for each role reversal.

OPTION 1: Starting Pitcher to Closer (**Graham Ashcraft - CIN**):
Initial Filters: Filter out all currently successful SPs (xERA < 4.35), all non-SPs, all pitchers with 3+ quality pitches, all pitchers with less than 1 QP, bad control (<100 location+), and all pitchers without an elite pitch (elite = pitch in the 95th percentile in Stuff+).

This filtering left 4 strong candidates; Graham Ashcraft, James McArthur, Griffin Jax, and Brock Stewart. McArthur was immediately dismissed, having too small a sample in MLB play. Griffin Jax was a strong candidate, but may be better suited as a SP (analysis below), and Brock Stewart (while a strong candidate), had characteristics that may make him a more effective RP/SP (ability to go more than 1 inning).

Graham Ashcraft stood out as the top candidate for a role change from a SP to a CL. Ashcraft has two pitches (other than fastball) with an above average Stuff+ metric (Slider and Cutter). His slider falls into the “elite” category of being in the 95th percentile for that pitch in all of baseball. He holds a slightly above average cutter. However, his third pitch (sinker) has one of the worst ratings in all of baseball. I am hypothesizing that once the sinker gets incorporated into his pitch mix, Ashcraft starts to get hit. This is reflected in his by-inning split stats. He holds a 2.08 ERA in the 1st inning of all starts, but holds a 7.36 ERA in the second inning, and a 4.74 ERA in the 3rd inning. This data implies that Ashcraft may be a really good one inning player, putting him in the ideal CL category.

OPTION 2: Relief Pitchers to Starting Pitcher Candidates (**JOHN WINCKOWSKI - BOS**):
Initial Filters: Filter out all pitchers that started more than 30% of their games played (any more is not a true RP), all pitchers with under 3 QPs, all pitchers with an xERA > 4.5 (little reasoning to think an unsuccessful pitcher will be successful in a more intense, starting role), anyone over the age of 30.

The above filters above, when applied, revealed 4 players of great interest; Josh Winckowski, Daniel Palencia, Beau Brieske, and Gabe Speier. Palencia was eliminated due to being a rookie, pitching only 28.1 innings in 2023. This player is still developing and there is not enough data to fairly evaluate him at this time. Gabe Speier was dismissed due to his career high IP in a season being 54.2 in 2023. An ideal candidate would have proven an ability to maintain success in a higher inning count over his career. Winckowski was chosen over Beau Brieske due to Brieske coming back from a right ulnar nerve entrapment after the all star break in 2023.

Josh Winckowski shares some metrics with some of the most successful SPs in the MLB. First, his pitch mix contains 4 pitches other than fastballs that are considered quality pitches (above league average). This pitch mix is essential for a SP to not be predictable when facing batters for a second and third time. In addition, he showed a solid xERA (3.71) over 84.1 IP in 2023, with above average location (101.9) and a good BB/9 (3.31). Finally, in split stats, he exhibits improvement in his pitches 26-50 showing he can handle at least 50 pitches per outing (3-4 innings), and is relatively untested in outings over 50 pitches. He has shown steady improvement from his first to second season. The only metric that does not support this change is the high WHIP of 1.423. At a minimum, Winckowski is worth a look as a SP in 2024.

OPTION 3: Relief Pitcher to Starting Pitcher Candidate (**Griffin Jax**):

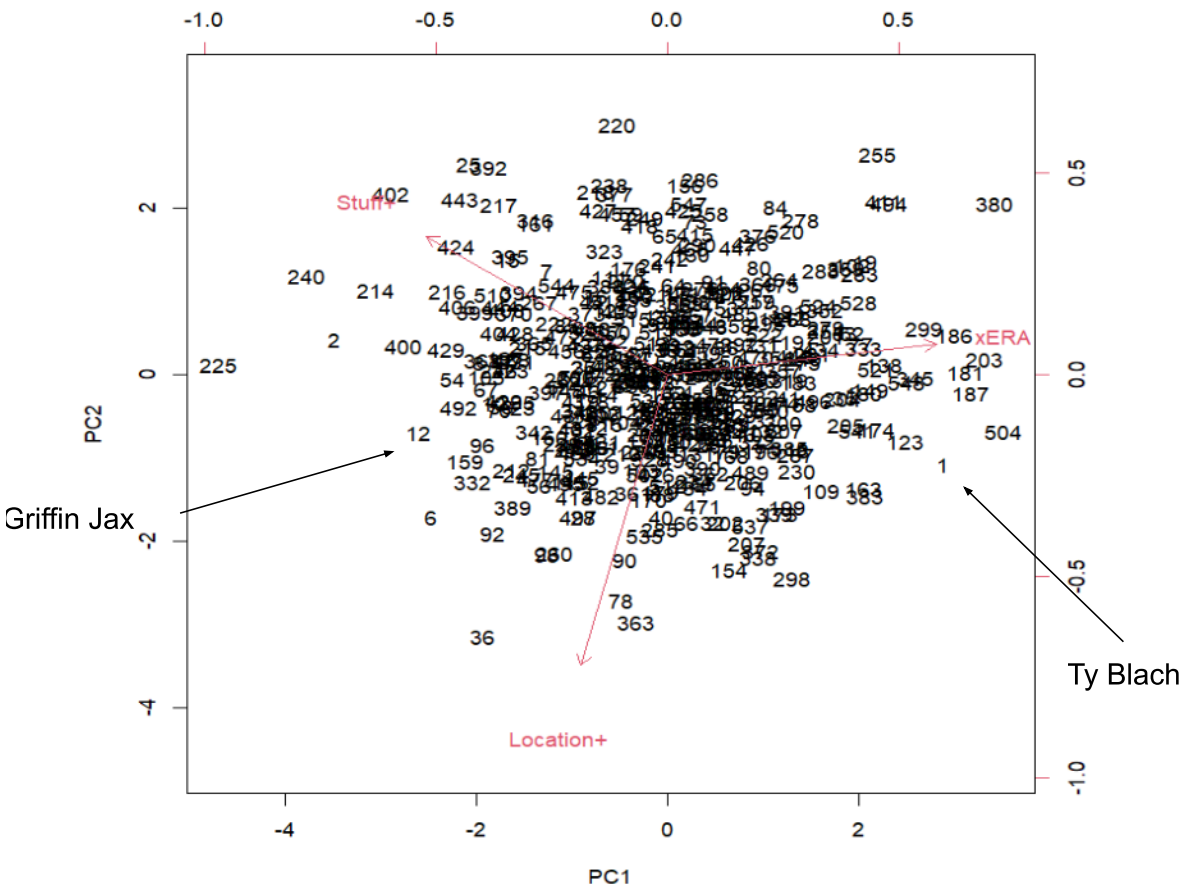
The initial filters used for this analysis were the filters used in option 1. Although that analysis was to identify potential closers, Jax stood out as a potential SP candidate. His pitch mix contains 4 quality pitches (includes an elite slider). He is currently a very successful RP, owning an xERA of 2.88 with a high K/9 of 9.37 and a low BB/9 of 2.62 (all great SP characteristics). He is just as effective against RHB as LHB according to split stats. Another appealing aspect of Griffin Jax is that he has proven well in a large workload of 65 innings last season. The jump from 65 IP to 120 IP is much more practical than transitioning someone from 30IP to a SP inning workload. In addition to being 29 years old (prime of career) with a consistent history, we can say that Jax is a good candidate to upgrade roles. The only negative to Jax is that his opposition OBP increased in pitches 26-50. However, this is based on 9 plate appearances (small sample size) and unusually high BABIP stat, and should not be extrapolated.

OPTION 4: Left-Handed Specialist (**Ty Blach**):

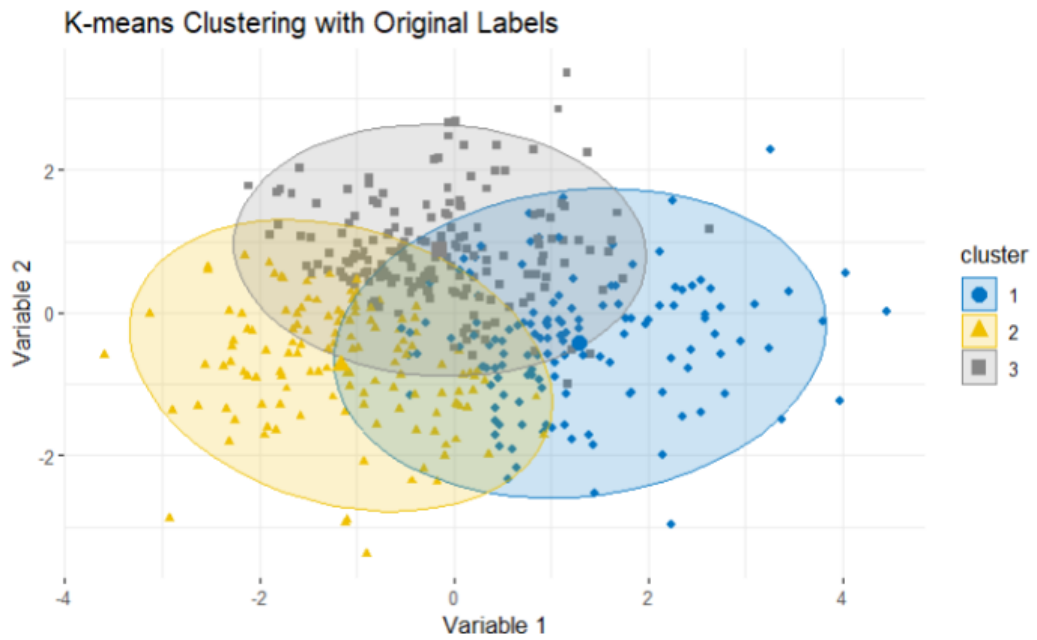
Ty Blach is an interesting case. He owns an abysmal 7.21 xERA and a 1.641 WHIP in 20 GP and 13 of those being starts. Ty is an older player (33 years old) and these stats show that his starting days in the MLB are over. However, looking deeper into Ty, there is a great use case for him. He owns the #1 ranked cutter ball according to Stuff+ in the MLB. His other two pitches (Curveball and Changeup) are among the worst in the league. One amazing pitch and two bad ones is not enough to survive long term in the MLB, but he might not have to. His split stats show that he is much more effective against LHB (opposing BA .269) than RHP (opposing BA .347). Ty Blach is also a left handed pitcher. He is a great fit to transition into a 1 out, left handed specialist similar to Tim Mayza on the Toronto Blue Jays.

Coding Methodology/Results:

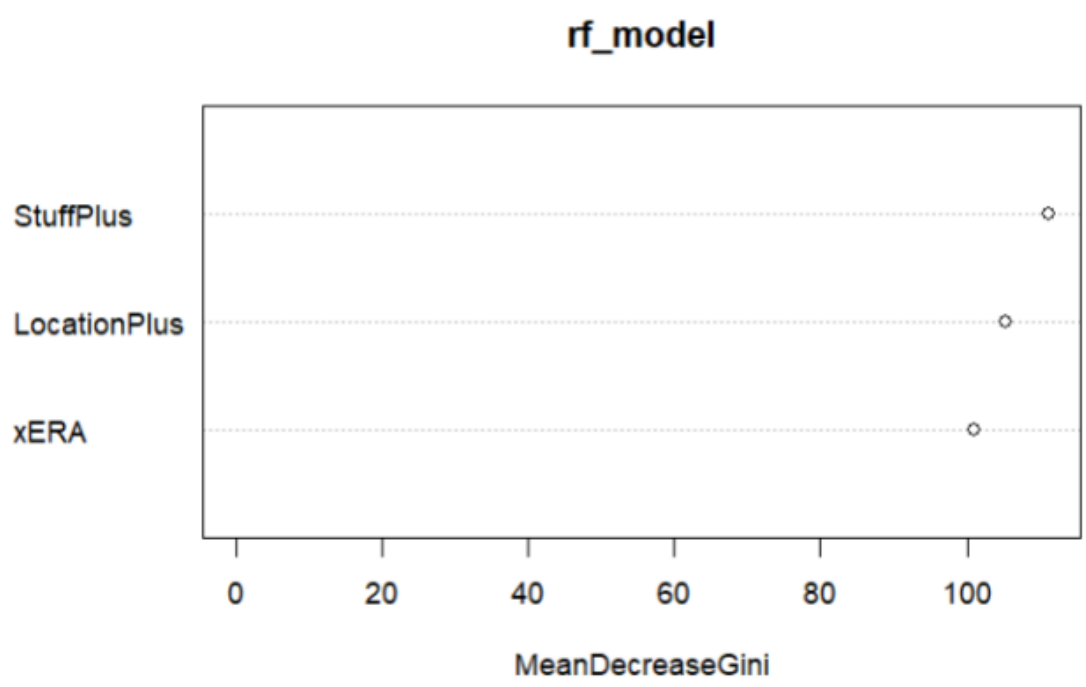
In order to assign a role swap to pitchers in our MLB pitcher data, we decided to use a few different evaluation methods. In order to run our evaluation methods, we first cleaned the data by removing missing variables and normalizing the data. Then, we also split the data into two sets (training and testing data) with an 80/20 split. The first evaluation method that we used was PCA evaluation. Within our PCA evaluation, we found some outliers. The outliers within the data are shown below.



The second evaluation method that we used was K-means clustering. The graph of K-means clustering is shown below:



After running our PCA evaluation, we were able to run K-means clustering on the data. We find in this case that the clusters are assumed to be Starting Pitchers in the blue, Relief Pitchers are assumed to be in the yellow, and Closing Pitchers are assumed to be in the gray.



Based on the random forest plot above, we were able to show that the most important variable in classifying the role of the pitcher is StuffPlus. (Stuff+), followed by LocationPlus (Location+), and xERA.

Lasly, we also used a Support Vector Machines (SVM analysis). Support Vector Machines (SVM) are a type of supervised machine learning algorithm that performs classification or regression tasks. The primary goal of SVM is to find a decision boundary (or hyperplane) that best separates data points belonging to different classes in a feature space.The results of our SVM procedure are shown below in a confusion matrix. This confusion matrix shows ~61% accuracy.

```
predictions_svm CL RP SP
             CL  2  0  0
             RP  8 34 13
             SP  1 20 32
[1] "Accuracy:  0.618181818181818"
```